This prototype contains a playable character capable of standing idly, walking, running, and jumping. The character was downloaded for free from the Unity Asset Store, and all of the animations were downloaded from Mixamo.

The character was animated using Unity's blend tree mechanic in order to connect all of the different animations in a configuration in which they can logically transition from one to another. The activation of the different animations as well as the movement of the character was done through Unity's New Input System, in which was created a "Move" action bound to the WASD keys through a 2D Vector, a "Run" action bound to the left control button, and a "Jump" action bound to the space bar. The POV of the player was created using Cinemachine's FreeLook camera by attaching it to the character object, allowing it to follow the character as it moves and providing the player with a third-person view of the character. The Cinemachine Deoccluder extension was also added to the FreeLook camera in order to help contain it within the greyboxing by preventing it from passing through walls and losing its view of the player character. The "Up" direction, ie. the direction in which the character moves when moving forward was also assigned to the direction in which the FreeLook camera is pointed in order to provide the player with an easy way to navigate the greyboxing environment.

The character feels satisfying to control for multiple reasons. First, its animations activate seamlessly depending on its current action, providing key visual feedback to the player. The Deoccluder extension attached to the FreeLook camera provides the player with a seamless view of the character, never allowing it to fall out of sight from the camera. The character's jumping and falling feel satisfying because of the selected gravity which provides the player with enough a reasonable amount of jump height while also bringing the character back down quickly, making it feel like it has some decent weight while also being agile.

By far the most difficult part of this project was getting the Unity Input System to function as intended. At first, the character would only be able to walk forward without possessing the ability to turn in any other direction. Then, even after it was capable of turning sideways, its internal compass was still stuck to the global “Up” direction instead of its local direction that it was facing. It took many hours of watching various Unity tutorials and coding scripts with C# in order to finally make the character walk in the direction that the Cinemachine FreeLook camera was facing instead of the global “Up” direction.